Introduction to the tidyverse

Scientific workflows: Tools and Tips 🛠️

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What is this lecture series?

Scientific workflows: Tools and Tips 🛠️

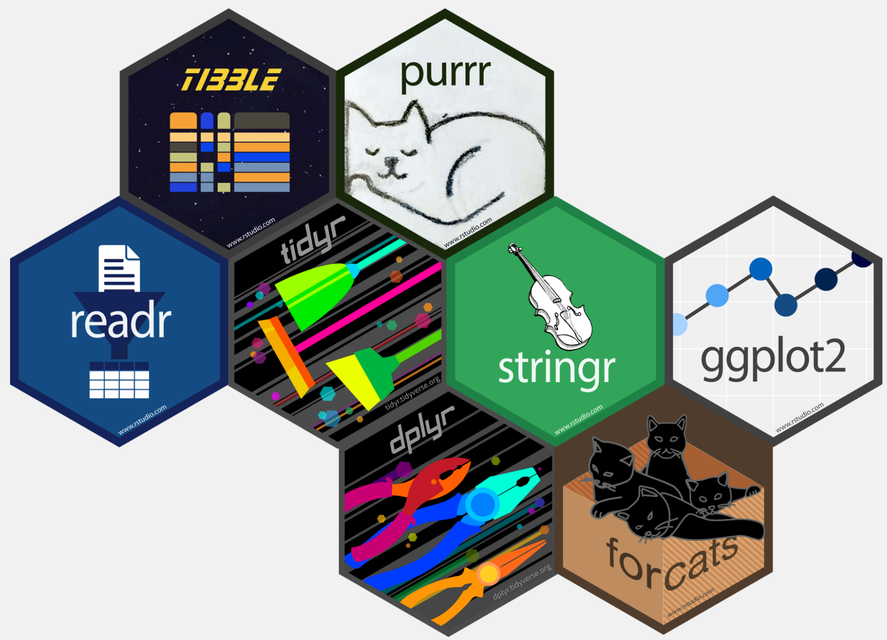
📅 Every 3rd Thursday 🕓 4-5 p.m. 📍 Webex

- One topic from the world of scientific workflows
- Material provided online
- If you don't want to miss a lecture
  - Subscribe to the mailing list

What is the tidyverse?

> The tidyverse is an opinionated **collection of R packages** designed for data science. All packages **share** an underlying **design philosophy, grammar, and data structures**.
>
> www.tidyverse.org

Tidyverse core packages



Why the tidyverse?

Basic idea: Make data analysis **efficient** and **intuitive**

- Write code that is easy to **read**, **write** and **maintain**
- More time for **data interpretation**
- Tidyverse is **actively developed** and has a **large community**.

Today

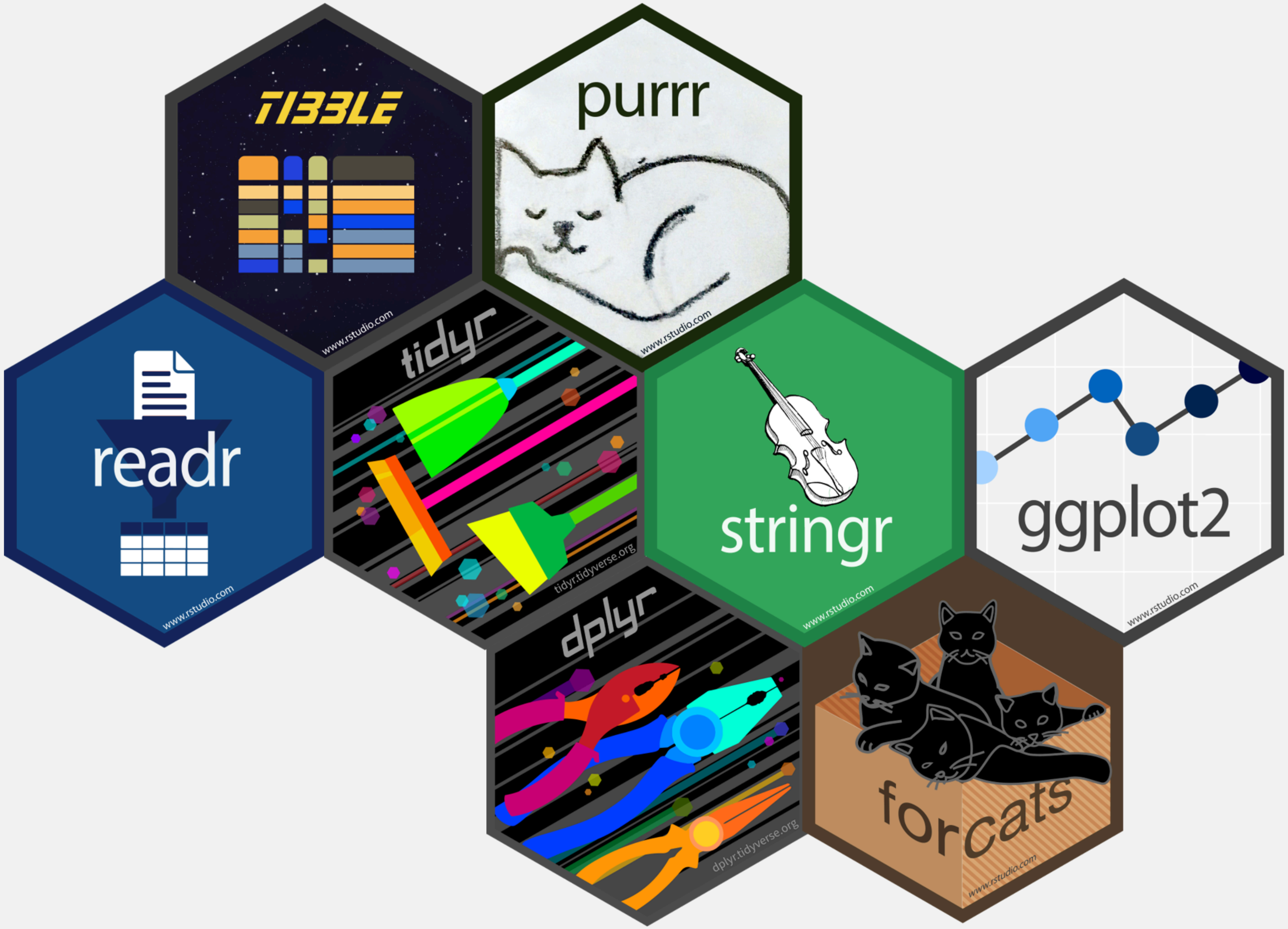
Overview of

- most important **packages and functions**
- how they work together in a **data analysis workflow**
- underlying **principles** of the tidyverse

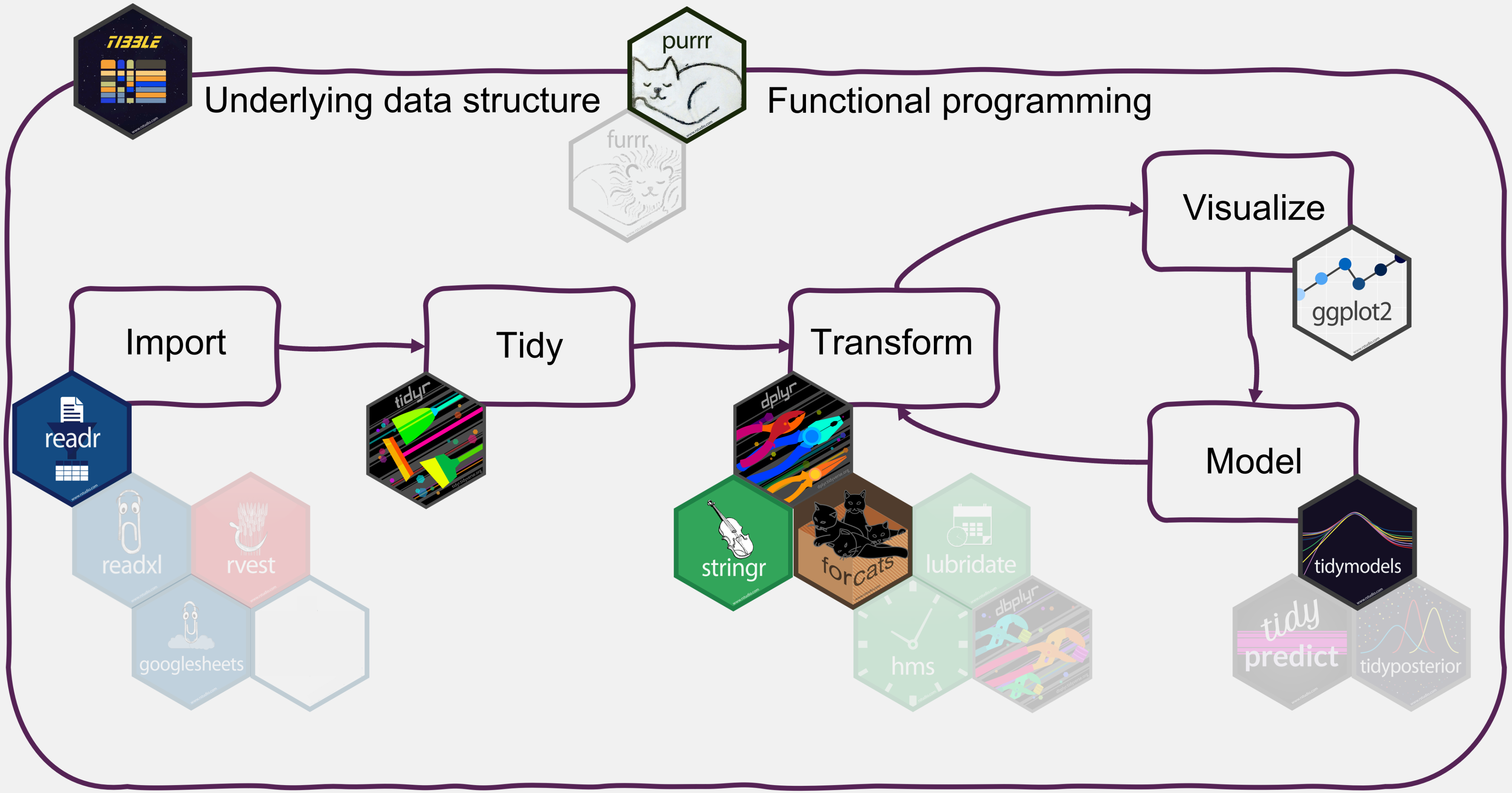
I can’t show you everything. But the tidyverse has an excellent documentation and cheatsheets.

Which tidyverse packages do you use?

Head over to the menti poll



Data analysis with the tidyverse



Getting the tidyverse

Install all tidyverse packages with

```r
install.packages("tidyverse")
```

Load all core tidyverse packages with

```r
library(tidyverse)
#> ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
#> ✓ dplyr     1.1.4     ✓ readr     2.1.5
#> ✓ forcats   1.0.0     ✓ stringr   1.5.1
#> ✓ ggplot2   3.5.1     ✓ tibble    3.2.1
#> ✓ lubridate 1.9.4     ✓ tidyr     1.3.1
#> ✓ purrr     1.0.2
#> ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
#> ✕ dplyr::filter() masks stats::filter()
#> ✕ dplyr::lag()    masks stats::lag()
#> ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

You can also load packages individually - But don't do both!

Non-core packages need to be loaded individually, e.g.

```r
library(readxl)
```

Tibbles from `tibble`

> The underlying data structure

Underlying data structure

What are tibbles?

Tibbles are

> a **modern reimagining of the data frame**. Tibbles are designed to be (as much as possible) **drop-in replacements** for data frames.
>
> (Wickham, Advanced R)

- Tibbles have the **same basic properties** as data frames
- Everything that you can do with data frames, you can do with tibbles
- Main advantage: Tibbles **print much nicer** to the console

Have a look at this book chapter for a full list of the differences between data frames and tibbles and the advantages of using tibbles.

How to create tibbles?

- Tidyverse functions return tibbles by default
- Create with `tibble` function (equivalent to `data.frame`)

How to create tibbles?

Creating a tibble

```
tibble(
  x = 1:26,
  y = letters,
  z = factor(LETTERS)
)
#> # A tibble: 26 × 3
#>        x y     z
#>    <int> <chr> <fct>
#>  1     1 a     A
#>  2     2 b     B
#>  3     3 c     C
#>  4     4 d     D
#>  5     5 e     E
#>  6     6 f     F
#>  7     7 g     G
#>  8     8 h     H
#>  9     9 i     I
#> 10    10 j     J
#> # ℹ 16 more rows
```

Creating a data frame

```
data.frame(
  x = 1:26,
  y = letters,
  z = factor(LETTERS)
)
#>     x y z
#> 1   1 a A
#> 2   2 b B
#> 3   3 c C
#> 4   4 d D
#> 5   5 e E
#> 6   6 f F
#> 7   7 g G
#> 8   8 h H
#> 9   9 i I
#> 10 10 j J
#> 11 11 k K
#> 12 12 l L
#> 13 13 m M
#> 14 14 n N
#> 15 15 o O
```

Import data with readr

> Read in your files

Most important readr functions

- `read_csv` to read **comma delimited** files
- `read_tsv` to read **tab delimited** files
- `read_delim` to read files with **any delimiter**

All `read_*` functions return a `tibble`.

Example with arctic temperature data

Data modified from `lterdatasampler`

```
arc_weather <- read_csv(file = "data/arc_weather.csv")
arc_weather
#> # A tibble: 365 × 3
#>    date       foolik_mean_temp_c poolik_mean_temp_c
#>    <date>                  <dbl>              <dbl>
#>  1 1988-06-01                8.4              11.8
#>  2 1988-06-02                6                 8.90
#>  3 1988-06-03                5.8               8.14
#>  4 1988-06-04                1.8               3.95
#>  5 1988-06-05                6.8               9.81
#>  6 1988-06-06                5.2               8.74
#>  7 1988-06-07                2.2               6.11
#>  8 1988-06-08                9.4              12.1
#>  9 1988-06-09               13.1              16.5
#> 10 1988-06-10               17.7              20.9
#> # ℹ 355 more rows
```

Advantages of `readr`

- Faster than `read.table`/`read.csv`
- Better defaults than base R (e.g. guessing of data types)
- Returns tibbles
- Tidyverse packages for other data types:
  - `readxl` for Excel files
  - `haven` for SPSS, SAS, and Stata files
  - `googlesheets4` for Google sheets
  - ...

Tidy data with `tidyr`

> Reorganize the data

Tidy

What is tidy data?

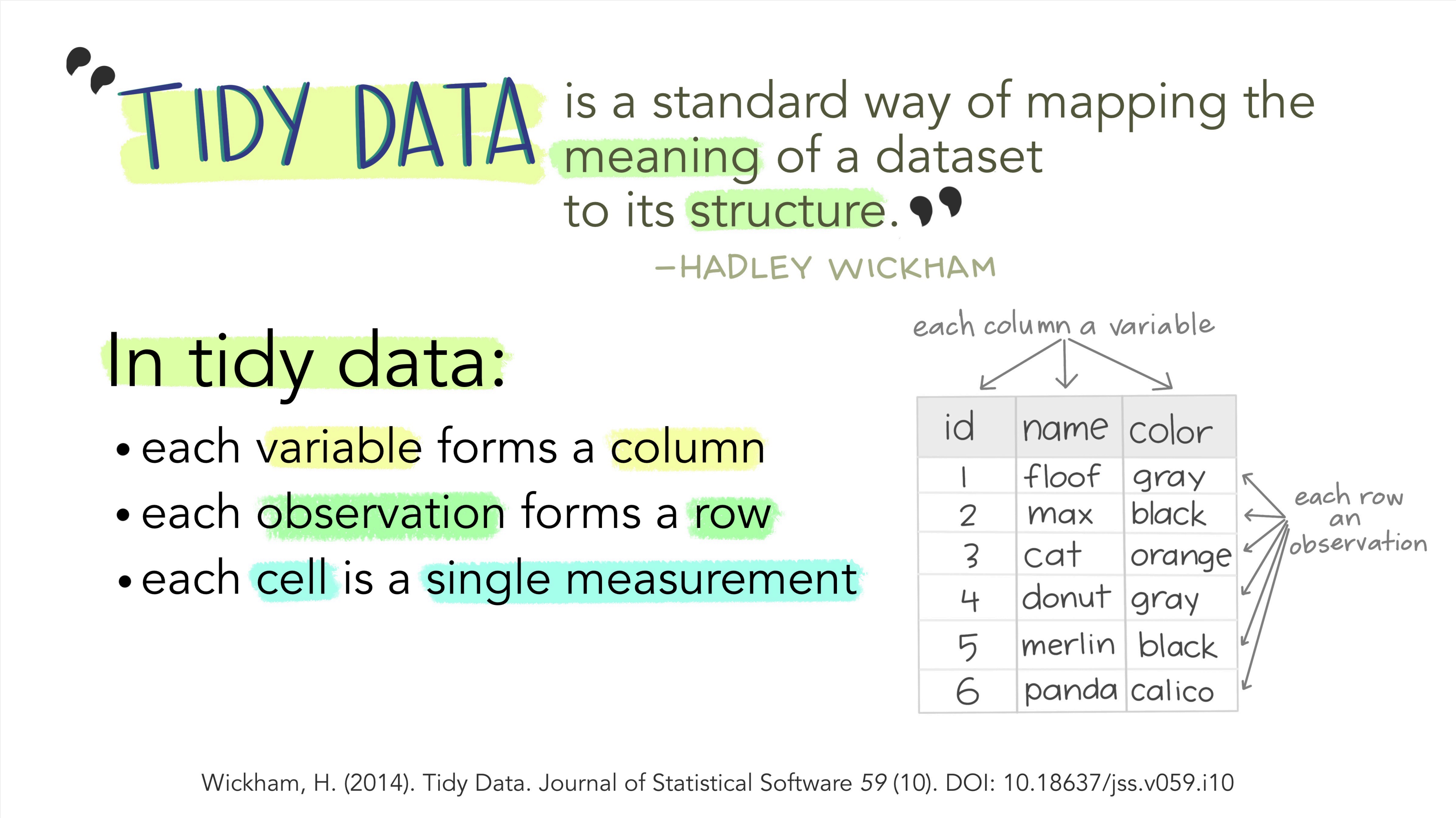


Illustration from the Openscapes blog *Tidy Data for reproducibility, efficiency, and collaboration* by Julia Lowndes and Allison Horst

What is non-tidy data?

Tidy ::: {.cell} ::: {.cell-output-display}

id	name	color
1	floof	gray
2	max	black
3	cat	orange
4	donut	gray
5	merlin	black
6	panda	calico

:::

Non-tidy ::: {.cell} ::: {.cell-output-display}

floof	max	cat	donut	merlin	panda
gray	black	orange	gray	black	calico

::: {.cell} ::: {.cell-output-display}

gray	black	orange	calico
floof	max	cat	panda
donut	merlin		

::: :::

:::

:::

:::

Advantages of tidy data

The main advantages of **tidy** data is that the `tidyverse` packages are built to work with it.

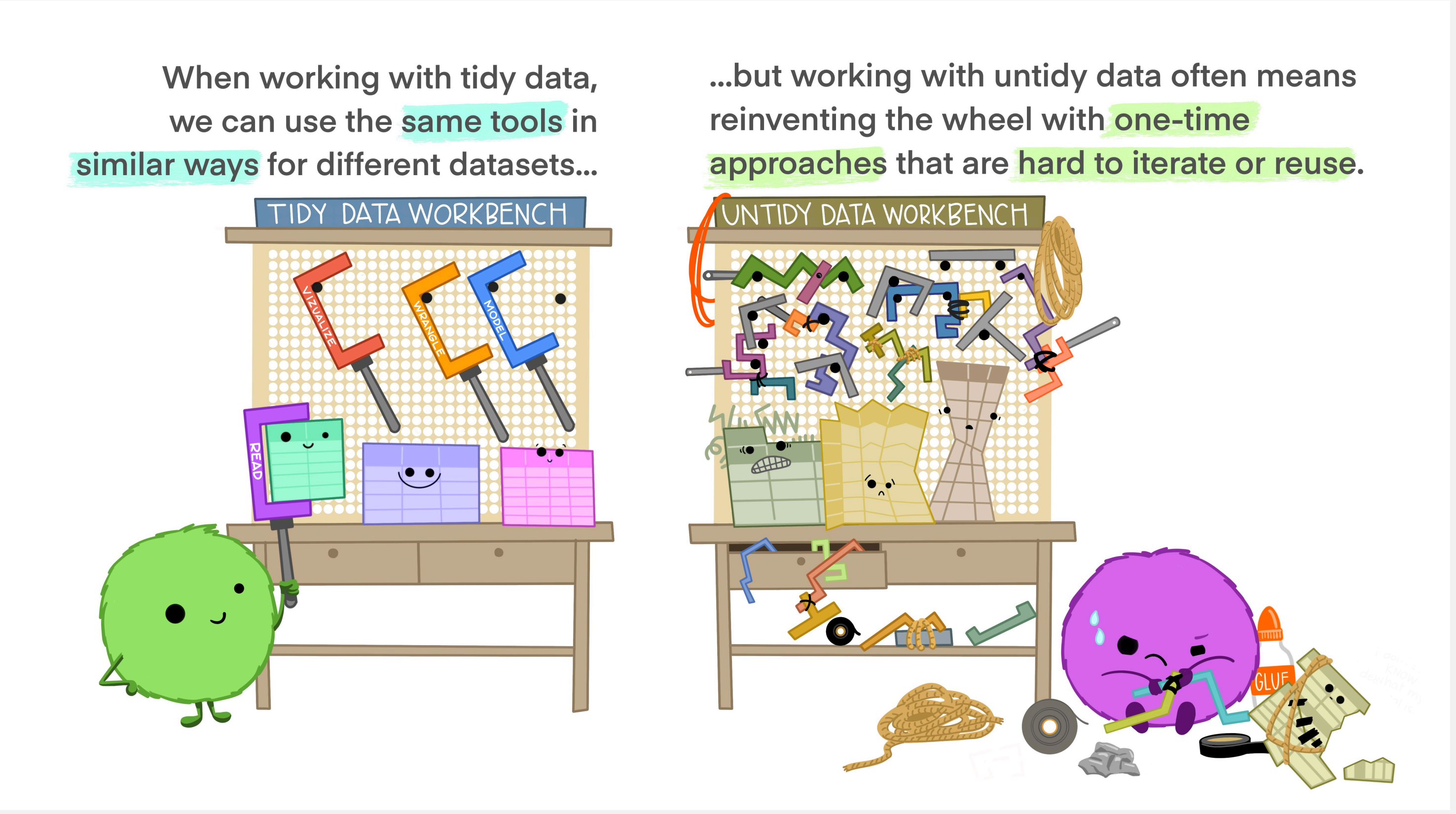


Illustration from the Openscapes blog *Tidy Data for reproducibility, efficiency, and collaboration* by Julia Lowndes and Allison Horst

Tidy data with tidyr

- `Tidyr` **reorganizes** the data but **does not change** values
- Most important functionality:
  - **Pivoting**: `pivot_longer` and `pivot_wider`
  - **Splitting** and **combining** columns: `separate_wider_delim/position/regex` and `unite`
  - Handle **missing values**: `drop_na`, `replace_na`, `complete`

Example

What is not tidy about our weather data?

```
arc_weather
#> # A tibble: 365 × 3
#>    date       foolik_mean_temp_c poolik_mean_temp_c
#>    <date>                  <dbl>              <dbl>
#>  1 1988-06-01                8.4              11.8
#>  2 1988-06-02                6                 8.90
#>  3 1988-06-03                5.8               8.14
#>  4 1988-06-04                1.8               3.95
#>  5 1988-06-05                6.8               9.81
#>  6 1988-06-06                5.2               8.74
#>  7 1988-06-07                2.2               6.11
#>  8 1988-06-08                9.4              12.1
#>  9 1988-06-09               13.1              16.5
#> 10 1988-06-10               17.7              20.9
#> # ℹ 355 more rows
```

- Each row has multiple observations
- Variables are split across multiple columns

Example

This can be solved with `pivot_longer`

```
pivot_longer(arc_weather,
  cols = c(foolik_mean_temp_c, poolik_mean_temp_c))
#> # A tibble: 730 × 3
#>    date       name               value
#>    <date>     <chr>              <dbl>
#>  1 1988-06-01 foolik_mean_temp_c  8.4
#>  2 1988-06-01 poolik_mean_temp_c 11.8
#>  3 1988-06-02 foolik_mean_temp_c  6
#>  4 1988-06-02 poolik_mean_temp_c  8.90
#>  5 1988-06-03 foolik_mean_temp_c  5.8
#>  6 1988-06-03 poolik_mean_temp_c  8.14
#>  7 1988-06-04 foolik_mean_temp_c  1.8
#>  8 1988-06-04 poolik_mean_temp_c  3.95
#>  9 1988-06-05 foolik_mean_temp_c  6.8
#> 10 1988-06-05 poolik_mean_temp_c  9.81
#> # ℹ 720 more rows
```

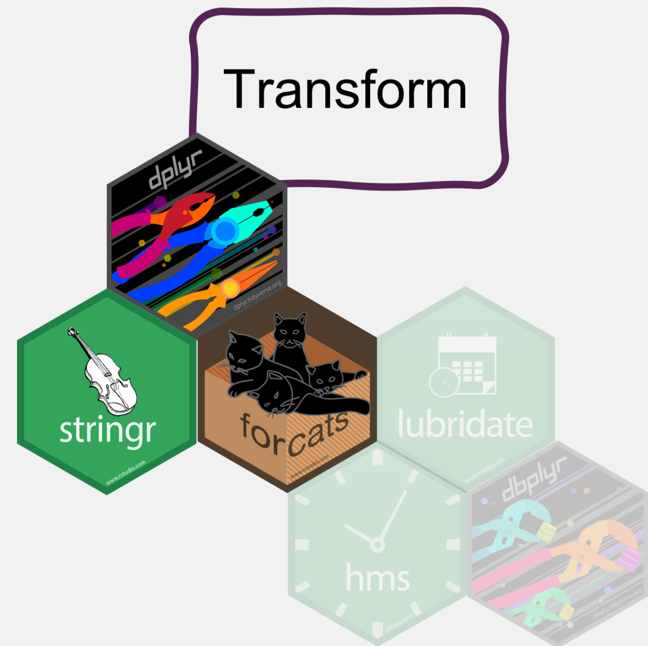
Example

You can also directly name the new columns:

```
arc_weather <- pivot_longer(arc_weather,
  cols = c(foolik_mean_temp_c, poolik_mean_temp_c),
  names_to = "station",
  values_to = "mean_temp_c")
arc_weather
#> # A tibble: 730 × 3
#>    date       station            mean_temp_c
#>    <date>     <chr>                    <dbl>
#>  1 1988-06-01 foolik_mean_temp_c        8.4
#>  2 1988-06-01 poolik_mean_temp_c       11.8
#>  3 1988-06-02 foolik_mean_temp_c        6
#>  4 1988-06-02 poolik_mean_temp_c        8.90
#>  5 1988-06-03 foolik_mean_temp_c        5.8
#>  6 1988-06-03 poolik_mean_temp_c        8.14
#>  7 1988-06-04 foolik_mean_temp_c        1.8
#>  8 1988-06-04 poolik_mean_temp_c        3.95
#>  9 1988-06-05 foolik_mean_temp_c        6.8
#> 10 1988-06-05 poolik_mean_temp_c        9.81
#> # ℹ 720 more rows
```

Transform data with `dplyr`

> How to actually change values



Transform data with `dplyr`

- Data cleaning, adding new columns, summarizing, …
- Depending on the data type in combination with
  - `stringr` for character columns
  - `lubridate` for date-time columns
  - `forcats` for factor columns

Data cleaning and filtering

`select` picks variables (columns) based on their names

```
select(arc_weather,
  date, mean_temp_c)
#> # A tibble: 730 × 2
#>    date       mean_temp_c
#>    <date>           <dbl>
#>  1 1988-06-01        8.4
#>  2 1988-06-01       11.8
#>  3 1988-06-02        6
#>  4 1988-06-02        8.90
#>  5 1988-06-03        5.8
#>  6 1988-06-03        8.14
#>  7 1988-06-04        1.8
#>  8 1988-06-04        3.95
#>  9 1988-06-05        6.8
#> 10 1988-06-05        9.81
#> # ℹ 720 more rows
```

`filter` picks observations (rows) based on their values

```
filter(arc_weather,
  station == "foolik_mean_temp_c")
#> # A tibble: 365 × 3
#>    date       station            mean_temp_c
#>    <date>     <chr>                    <dbl>
#>  1 1988-06-01 foolik_mean_temp_c         8.4
#>  2 1988-06-02 foolik_mean_temp_c         6
#>  3 1988-06-03 foolik_mean_temp_c         5.8
#>  4 1988-06-04 foolik_mean_temp_c         1.8
#>  5 1988-06-05 foolik_mean_temp_c         6.8
#>  6 1988-06-06 foolik_mean_temp_c         5.2
#>  7 1988-06-07 foolik_mean_temp_c         2.2
#>  8 1988-06-08 foolik_mean_temp_c         9.4
#>  9 1988-06-09 foolik_mean_temp_c        13.1
#> 10 1988-06-10 foolik_mean_temp_c        17.7
#> # ℹ 355 more rows
```

Combine `filter` with other packages

`lubridate` to filter by month

```
# Filter rows from May
filter(arc_weather,
  month(date) == 5)
#> # A tibble: 62 × 3
#>    date       station             mean_temp_c
#>    <date>     <chr>                     <dbl>
#>  1 1989-05-01 foolik_mean_temp_c       -3.3
#>  2 1989-05-01 poolik_mean_temp_c       -0.312
#>  3 1989-05-02 foolik_mean_temp_c       -0.8
#>  4 1989-05-02 poolik_mean_temp_c        2.20
#>  5 1989-05-03 foolik_mean_temp_c        0.5
#>  6 1989-05-03 poolik_mean_temp_c        2.59
#>  7 1989-05-04 foolik_mean_temp_c       -2.1
#>  8 1989-05-04 poolik_mean_temp_c        0.457
#>  9 1989-05-05 foolik_mean_temp_c       -2
#> 10 1989-05-05 poolik_mean_temp_c        1.19
#> # ℹ 52 more rows
```

`stringr` to filter characters

```
# Filter rows where the station contains "foolik"
filter(arc_weather,
  str_detect(station, "foolik"))
#> # A tibble: 365 × 3
#>    date       station             mean_temp_c
#>    <date>     <chr>                     <dbl>
#>  1 1988-06-01 foolik_mean_temp_c          8.4
#>  2 1988-06-02 foolik_mean_temp_c          6
#>  3 1988-06-03 foolik_mean_temp_c          5.8
#>  4 1988-06-04 foolik_mean_temp_c          1.8
#>  5 1988-06-05 foolik_mean_temp_c          6.8
#>  6 1988-06-06 foolik_mean_temp_c          5.2
#>  7 1988-06-07 foolik_mean_temp_c          2.2
#>  8 1988-06-08 foolik_mean_temp_c          9.4
#>  9 1988-06-09 foolik_mean_temp_c         13.1
#> 10 1988-06-10 foolik_mean_temp_c         17.7
#> # ℹ 355 more rows
```

Change and add values

Add columns with `mutate`

```
# Add column with temperature in K
mutate(arc_weather,
  mean_temp_k = mean_temp_c + 274.15)
#> # A tibble: 730 × 4
#>    date       station            mean_temp_c mean_temp_k
#>    <date>     <chr>                    <dbl>       <dbl>
#>  1 1988-06-01 foolik_mean_temp_c        8.4         283.
#>  2 1988-06-01 poolik_mean_temp_c       11.8         286.
#>  3 1988-06-02 foolik_mean_temp_c        6           280.
#>  4 1988-06-02 poolik_mean_temp_c        8.90        283.
#>  5 1988-06-03 foolik_mean_temp_c        5.8         280.
#>  6 1988-06-03 poolik_mean_temp_c        8.14        282.
#>  7 1988-06-04 foolik_mean_temp_c        1.8         276.
#>  8 1988-06-04 poolik_mean_temp_c        3.95        278.
#>  9 1988-06-05 foolik_mean_temp_c        6.8         281.
#> 10 1988-06-05 poolik_mean_temp_c        9.81        284.
#> # ℹ 720 more rows
```

Change and add values

Change the values of existing columns with `mutate`

```
# Remove _mean_temp_c from station
arc_weather <- mutate(arc_weather,
                        station = str_remove(station, "_mean_temp_c"))
arc_weather
#> # A tibble: 730 × 3
#>    date       station mean_temp_c
#>    <date>     <chr>         <dbl>
#>  1 1988-06-01 foolik         8.4
#>  2 1988-06-01 poolik        11.8
#>  3 1988-06-02 foolik         6
#>  4 1988-06-02 poolik         8.90
#>  5 1988-06-03 foolik         5.8
#>  6 1988-06-03 poolik         8.14
#>  7 1988-06-04 foolik         1.8
#>  8 1988-06-04 poolik         3.95
#>  9 1988-06-05 foolik         6.8
#> 10 1988-06-05 poolik         9.81
#> # ℹ 720 more rows
```

Summarize data

`summarize` will **collapse the data** to a **single row**

```
# Calculate the overall mean temperature
summarize(arc_weather,
            overall_mean = mean(mean_temp_c, na.rm = TRUE))
#> # A tibble: 1 × 1
#>   overall_mean
#>          <dbl>
#> 1        -7.10
```

Summarizing data by group

`summarize` in combination with `group_by` will **collapse the data** to a **single row per group**

```
# Calculate the mean temperature for each station
arc_weather_group <- group_by(arc_weather, station)
summarize(arc_weather_group,
  overall_mean = mean(mean_temp_c, na.rm = TRUE))
#> # A tibble: 2 × 2
#>   station overall_mean
#>   <chr>          <dbl>
#> 1 foolik         -8.60
#> 2 poolik         -5.60
```

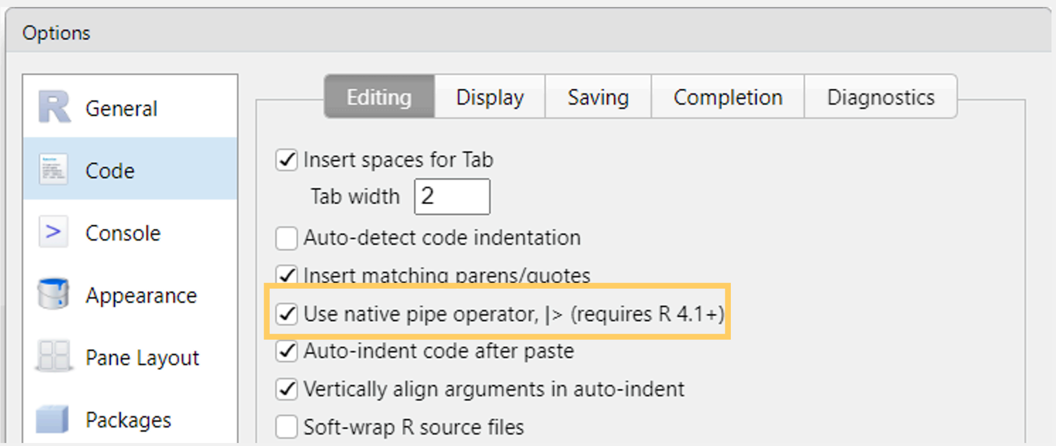
- `group_by` can be used for any dplyr function to perform the operation by group

The pipe |>

Data transformation often requires **multiple operations** in sequence.

The pipe operator `|>` helps to keep these operations clear and readable.

- You may also see `%>%` from the `magrittr` package (tidyverse)
- Turn on the native R pipe `|>` in **Tools -> Global Options -> Code**



The pipe |>

Let's look at an example without the pipe:

```
arc_weather_group <- group_by(arc_weather, station)
summarize(arc_weather_group,
          overall_mean = mean(mean_temp_c, na.rm = TRUE))
```

The pipe operator makes it very easy to combine these operations

```
arc_weather |>
  group_by(station) |>
  summarize(overall_mean = mean(mean_temp_c, na.rm = TRUE))
```

You can read from top to bottom and interpret the |> as an "*and then do*".

The pipe |>

But what is happening?

The pipe is "pushing" the result of one line into the first argument of the function from the next line.

```
arc_weather |>
  group_by(station)

# instead of
group_by(arc_weather, station)
```

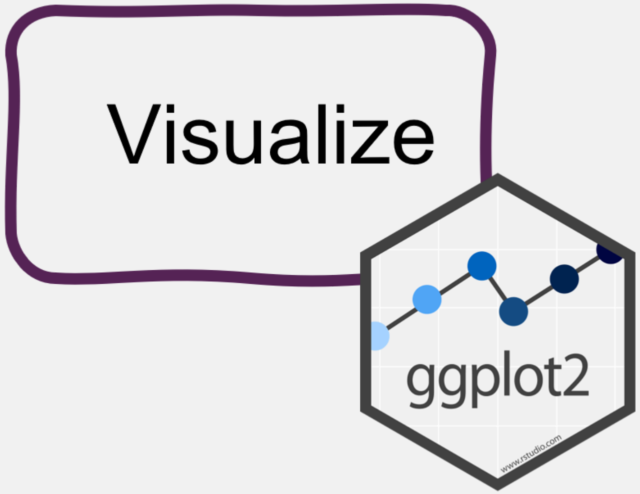
Piping works perfectly with the `tidyverse` functions because they are designed to **return a tibble** and **take a tibble as first argument**.

Tip

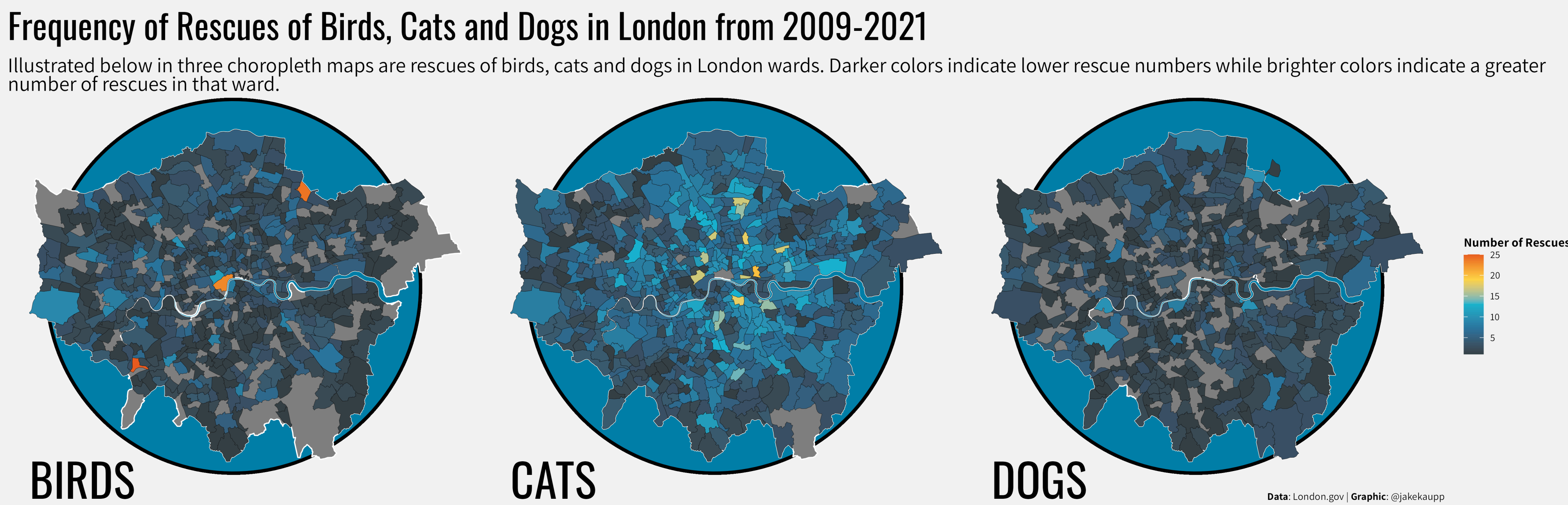
Use the keyboard shortcut `Ctrl/Cmd + Shift + M` to insert `|>`

Visualize data with `ggplot2`

> Quick exploratory plots and data masterpieces

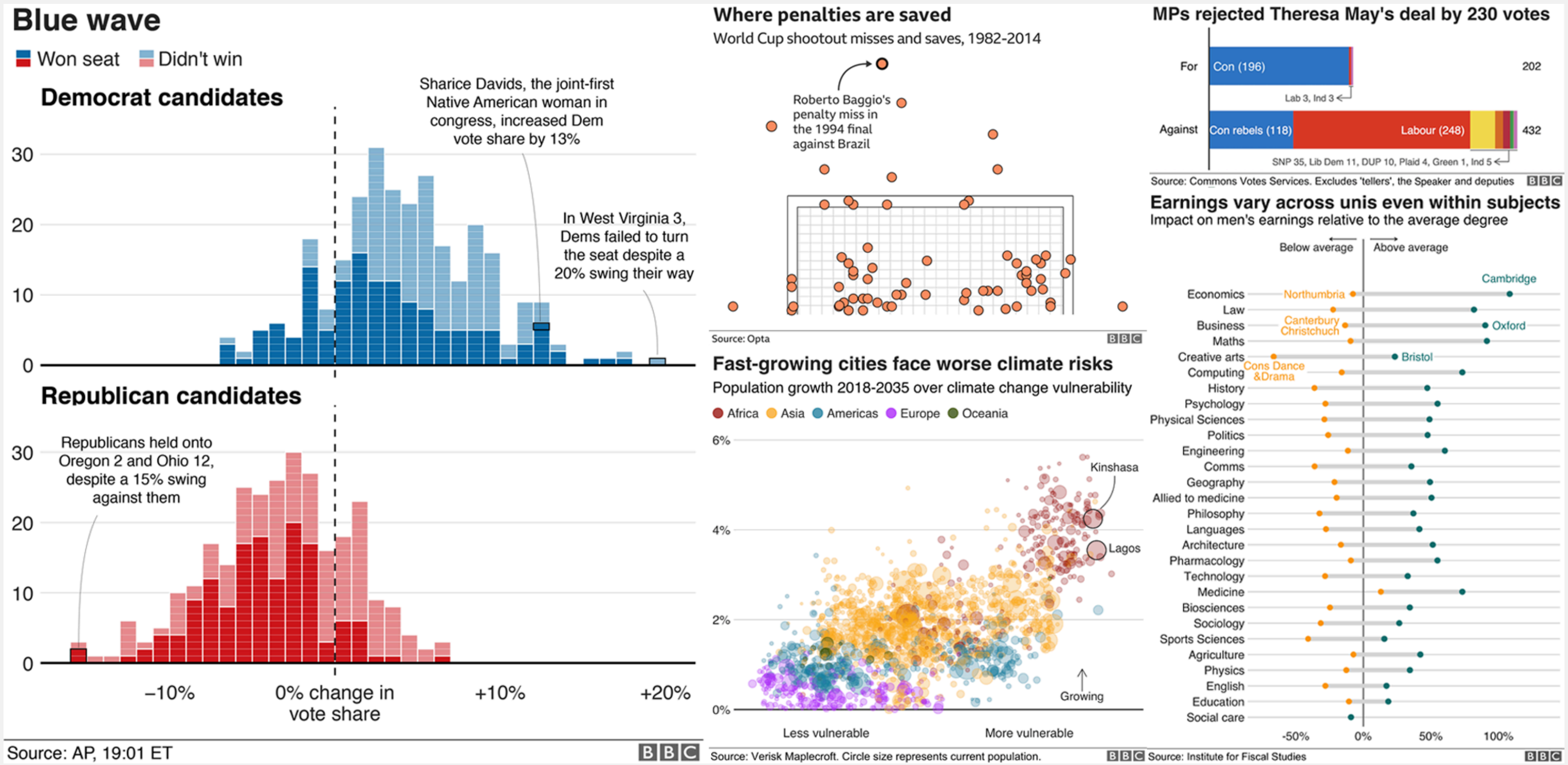


A ggplot showcase



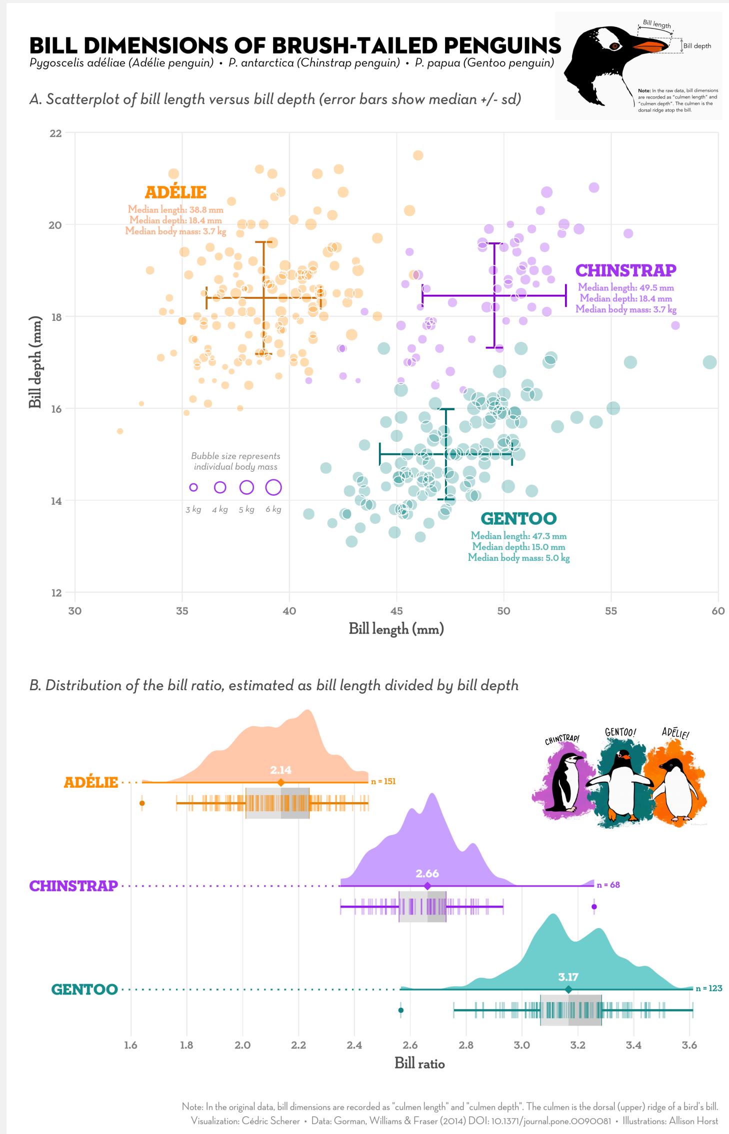
Visualization by Jake Kaupp, code available on Github

A ggplot showcase



Visualizations produced by the BBC News data team

A ggplot showcase



Visualization by Cédric Scherer, code available on Github

Advantages of ggplot

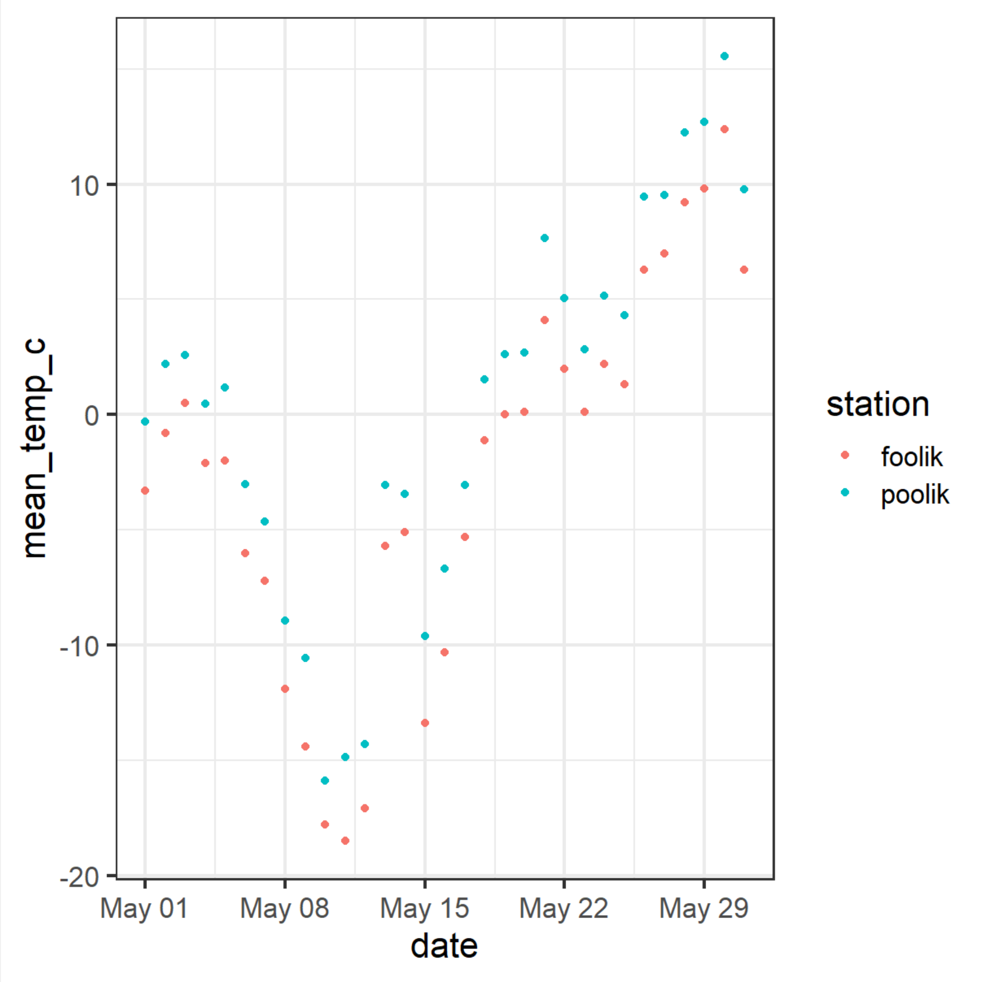
- **Consistent** grammar
- **Flexible** structure allows you to produce any type of plots
- Highly **customizable appearance** (themes)
- Many **extension packages** that provide additional plot types, themes, colors, animation, …
  - Find a list of ggplot extensions here
- Perfect package for **exploratory data analysis** and **beautiful plots**

Basic ggplot

Basic idea: Stack layers of the plot on top of each other with +

```
arc_weather_may <- filter(arc_weather,
  month(date) == 5)
ggplot(data = arc_weather_may,
  aes(
    x = date,
    y = mean_temp_c,
    color = station
)) +
  geom_point()
```

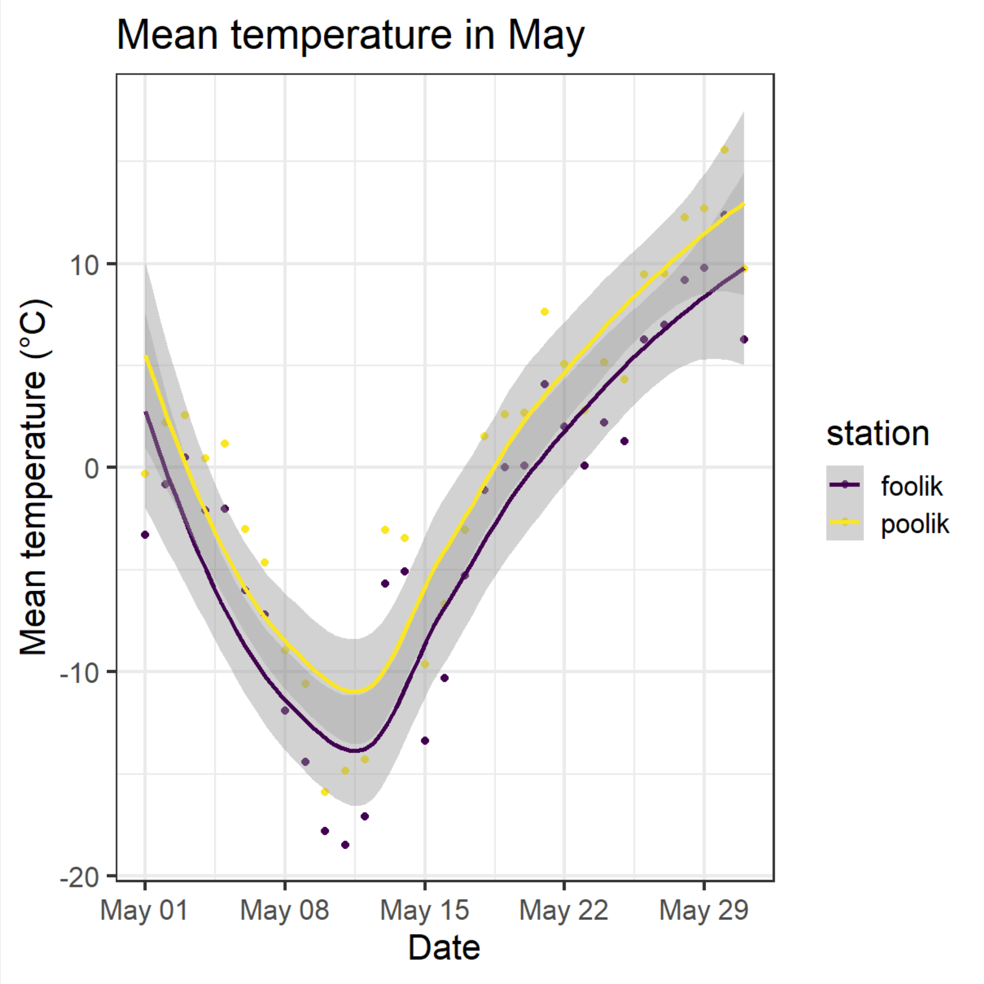
- **aes**thetics define how data variables are mapped onto plot properties
- **geoms** define the type of plot (e.g. points, lines, bars, ...)



Basic ggplot

- Gplot offers many more customization options and layers
- Works well with the pipe `|>`

```r
arc_weather |>
  filter(month(date) == 5) |>
  ggplot(aes(
    x = date,
    y = mean_temp_c,
    color = station)) +
  geom_point() +
  geom_smooth() +
  labs(
    title = "Mean temperature in May",
    x = "Date",
    y = "Mean temperature (°C)") +
  scale_color_viridis_d()
```



Functional programming with purrr

Functional programming

Functional programming with `purrr`

- Apply functions to multiple elements of a list/vector/…
  - Replace e.g. for loops/apply-functions
  - Often `purrr` functions are more intuitive
  - See here for a full comparison between base R functions and purr functions
- Most important function: `map`
  - Comes in different versions, depending on the input and the desired output

A simple example

Draw 4 numbers from normal distribution with means 1, 2, 3, 4.

Do it by hand

```
# Do it by hand
set.seed(123)
rnorm(n = 4, mean = 1, sd = 1)
#> [1] 0.4395244 0.7698225 2.5587083 1.0705084
rnorm(n = 4, mean = 2, sd = 1)
#> [1] 2.1292877 3.7150650 2.4609162 0.7349388
rnorm(n = 4, mean = 3, sd = 1)
#> [1] 2.313147 2.554338 4.224082 3.359814
rnorm(n = 4, mean = 4, sd = 1)
#> [1] 4.400771 4.110683 3.444159 5.786913
```

Use purrr

```
# Use purrr
set.seed(123)
means <- 1:4
samples <- map(means, rnorm, n = 4, sd = 1)
str(samples)
#> List of 4
#>  $ : num [1:4] 0.44 0.77 2.56 1.07
#>  $ : num [1:4] 2.129 3.715 2.461 0.735
#>  $ : num [1:4] 2.31 2.55 4.22 3.36
#>  $ : num [1:4] 4.4 4.11 3.44 5.79
```

Different versions of the `map` function

These are just some examples, there are many more options:

Output	Input	Purr function
List	1 Vector	map
List	2 Vectors	map2
Vector of desired type	1 Vector	map_lgl
Vector of desired type	2 Vectors	map2_lgl
Data frame	1 Vector	map_df
Data frame	2 Vectors	map2_df

A more complex example

Read in and analyse multiple files with the same structure at once.

```
# Step 1: List all files in your data directory
paths <- list.files("data")
```

```
#> [1] "foolik.csv" "poolik.csv"
```

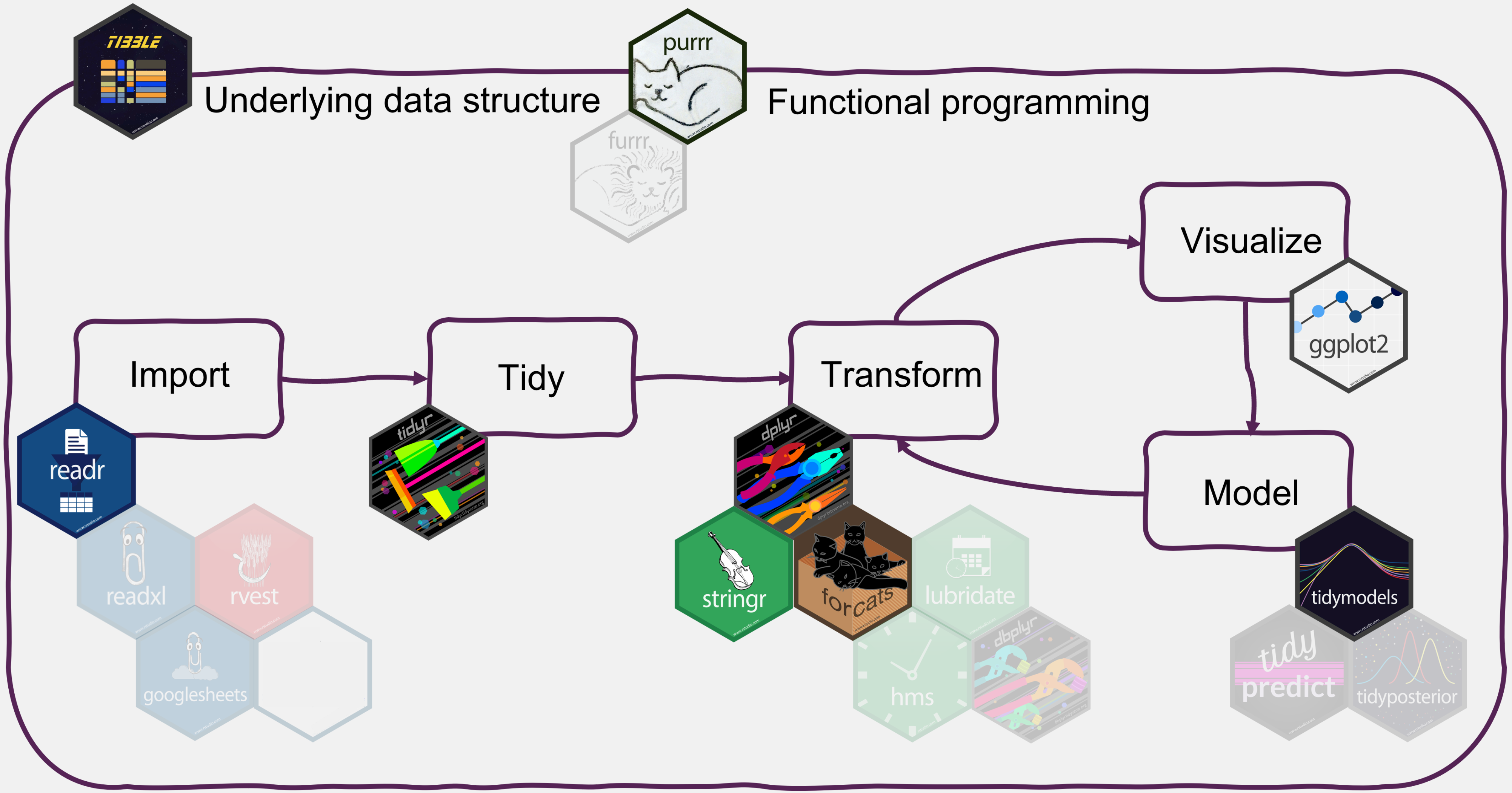
```
# Step 2: Read all files into a list of tibbles
arc_weather_files <- map(paths, read_csv)
arc_weather_files
#> [[1]]
#> # A tibble: 365 × 3
#>    date       station mean_temp_c
#>    <date>     <chr>         <dbl>
#>  1 1988-06-01 foolik          8.4
#>  2 1988-06-02 foolik          6
#>  3 1988-06-03 foolik          5.8
#>  4 1988-06-04 foolik          1.8
#>  5 1988-06-05 foolik          6.8
#>  6 1988-06-06 foolik          5.2
#>  7 1988-06-07 foolik          2.2
#>  8 1988-06-08 foolik          9.4
#>  9 1988-06-09 foolik         13.1
#> 10 1988-06-10 foolik         17.7
#> # ℹ 355 more rows
#>
#> [[2]]
```

A more complex example

Read in and analyse multiple files with the same structure at once.

```
# Step 3: Apply a linear model to each station file,
# get the model summary of each model and extract the coefficients
arc_weather_files |>                            # Take the data
  map(\(x) lm(mean_temp_c ~ date, data = x)) |> # Apply the same model to every tibbel in the list
  map(summary) |>                               # Get the model summary of both models
  map_dbl("r.squared")                          # Extract the r-squared from the model summaries
#> [1] 0.2018279 0.2000210
```

Summary



TIBBLE
Underlying data structure
purrr
furrr
Functional programming
Visualize
ggplot2
Import
Tidy
Transform
Model
readr
readxl
rvest
googlesheets
tidyr
dplyr
stringr
forcats
lubridate
hms
dbplyr
tidymodels
tidy predict
tidyposterior

Summary

Tidyverse makes data analysis

- efficient
- intuitive
- consistent

Tidyverse code is more **readable** and **easier to write & maintain**.

This facilitates

📚 Learning

🤝 Collaboration

🔄 Reproducible research

Summary

Where to get started with the tidyverse?

- Check out the lecture website for links to documenation, cheatsheets and book

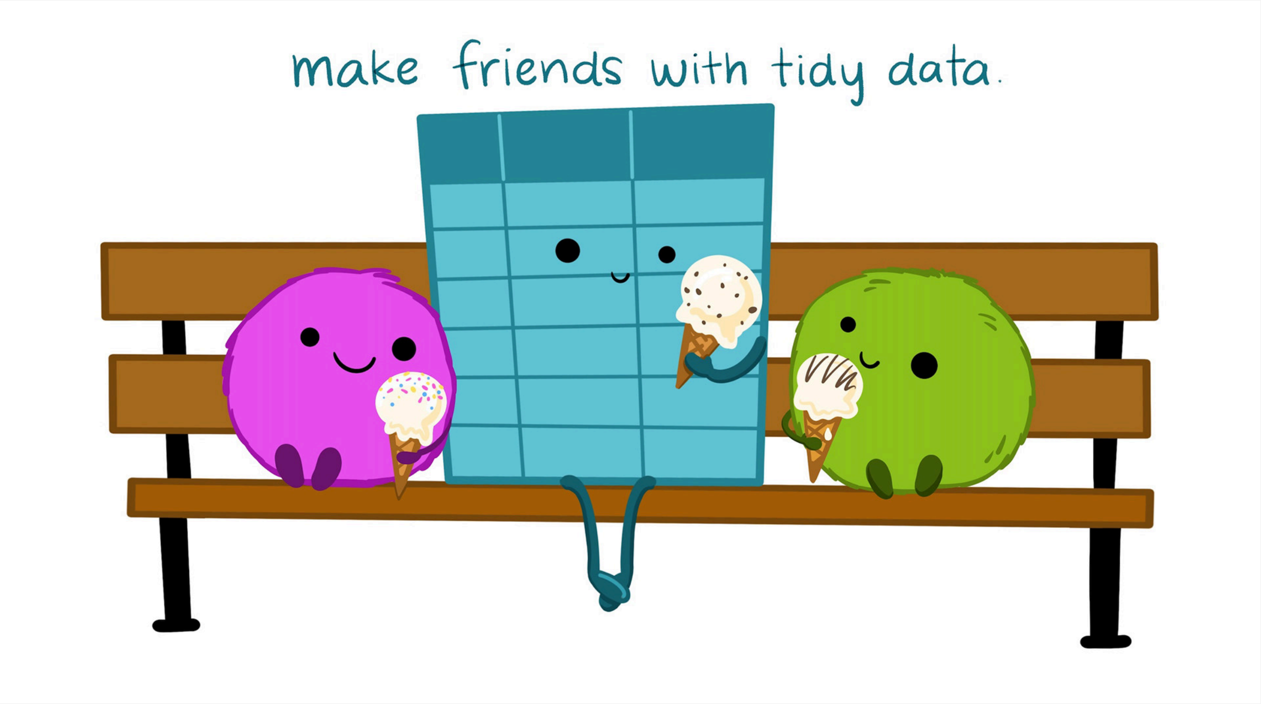


Illustration from the Openscapes blog *Tidy Data for reproducibility, efficiency, and collaboration* by Julia Lowndes and Allison Horst

Next lecture

Semester break in March!

Topic t.b.a.

📅 18th April 🕓 4-5 p.m. 📍 Webex

🔔 Subscribe to the mailing list

📧 For topic suggestions and/or feedback send me an email

Thank you for your attention :)

Questions?

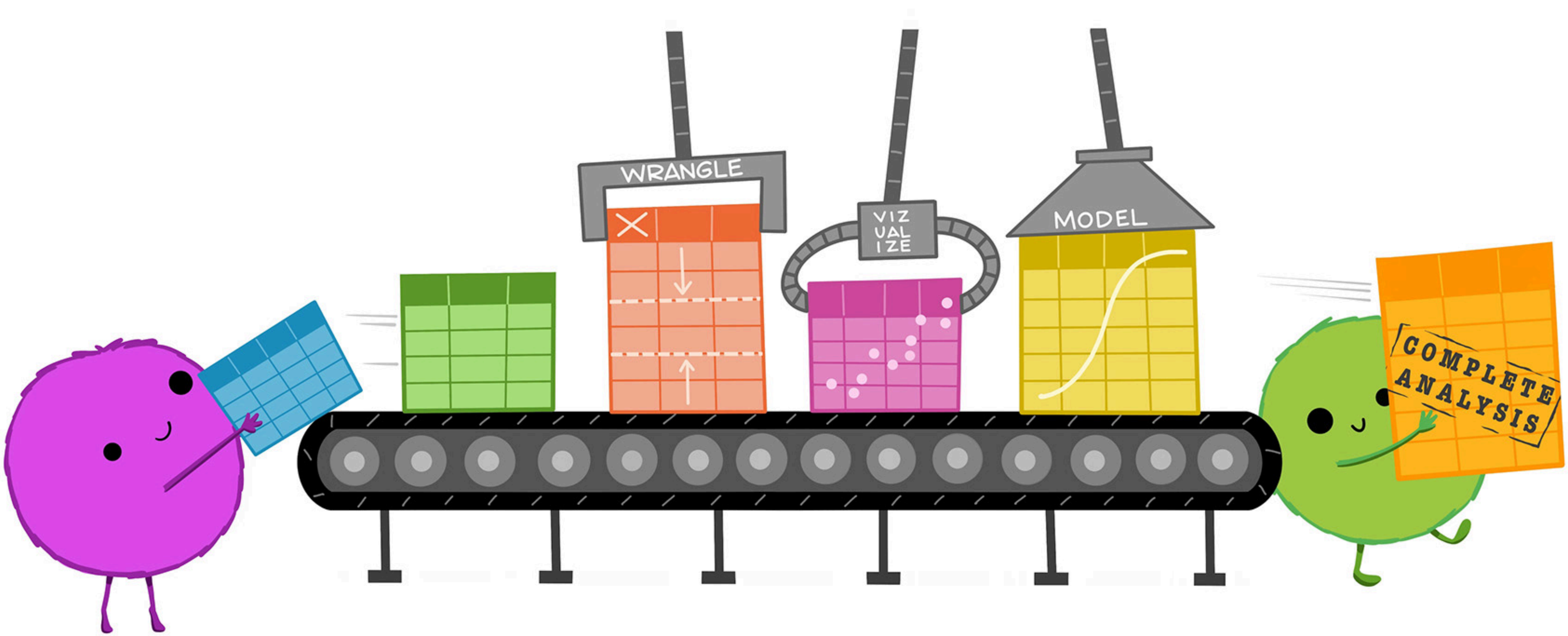


Illustration from the Openscapes blog *Tidy Data for reproducibility, efficiency, and collaboration* by Julia Lowndes and Allison

References

- Tidyverse website: find links to all package documentations
- Cheatsheets
- R for Data Science book: Learn data analysis with the tidyverse